



ABES ENGINEERING COLLEGE, GHAZIABAD (AUTONOMOUS)

Mid Term-I (Session: 2025-26)

Roll No.:

[Redacted Roll Number]

Course Name: B.Tech.

Subject Name: Fundamentals of Linear Algebra and Statistics

Subject Code: 25AS102

Max. Marks: 15

Time: 75 Minutes

Semester: 1st

Note:

1. Keeping Mobile Phone/Any Electronic Gadgets, Even in OFF position is treated as UFM
2. Don't Write Anything on Question Paper
3. Attempt All Questions
4. Assume Any Missing Data if required

SECTION-A (Maximum Knowledge Level: K2)				
Q. No.	Question	CO	KL	PI
1. Attempt All Questions (1 x 5)				
(a)	Find the sum and product of the eigen values of the matrix $\begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$.	CO3	K2	2.1.3
(b)	State Cayley Hamilton Theorem.	CO3	K1	2.1.3
(c)	Determine 'b' such that the system of homogeneous equations $2x + y + 2z = 0, x + y + 3z = 0, 4x + 3y + bz = 0$ has a nontrivial solution.	CO3	K2	2.1.3
(d)	The set of all second degree polynomial is not a vector space under the vector addition and scalar multiplication over R. Why?	CO1	K2	2.4.2
(e)	Define Subspace.	CO1	K1	2.4.1
SECTION-B (Minimum Knowledge Level: K2)				
2. Attempt Any two Question (Same K Levels) (2 x 2)				
(a)	Reduce the matrix A to normal form and hence find its rank, where $A = \begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 4 \\ 1 & 2 & 3 & 4 \\ -1 & -2 & 6 & -7 \end{bmatrix}$	CO3	K3	2.1.3
(b)	Employing elementary transformation find the inverse of the matrix $A = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 1 \\ 5 & 2 & -3 \end{bmatrix}$	CO3	K2	2.1.3
I	For what values of 'a' and 'b' the following system of linear equations: $x + 2y + 3z = 6, x + 3y + 5z = 9, 2x + 5y + az = b$, have (i) No solution (ii) Unique solution (iii) Infinitely many solutions	CO3	K3	2.1.3 5.2.2

SECTION-C (Minimum Knowledge Level: K3)

3. Attempt Any one Question

(Same K Levels) (3 x 1)

(a)	Find the eigen values and the corresponding eigen vectors of $A = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{bmatrix}$.	CO3	K3	2.1.3
(b)	Verify Cayley Hamilton Theorem for the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ and hence compute A^{-1} .	CO3	K3	2.1.3

4. Attempt Any one Question

(Same K Levels) (3 x 1)

(a)	Let $V = R^3$ be a vector space and $W = \{(x, y, z): x - 3y + 4z = 0\}$. Prove that W is a subspace of V over R.	CO1	K3	2.4.2 2.1.3
(b)	Prove that the set of all vectors in a plane (R^2) over the field of real numbers is a vector space with respect to vector addition and scalar multiplication.	CO1	K3	2.4.1 2.4.4

CO Course Outcomes mapped with respective question

KL Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create



ABES ENGINEERING COLLEGE, GHAZIABAD (AUTONOMOUS)

Mid Term-II (Session: 2025-26)

Roll No.:

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Course Name: B.Tech.

Subject Name: Fundamentals of Linear Algebra and Statistics

Subject Code: 25AS102

Max. Marks: 15

Time: 75 Minutes

Semester: I

Note:

1. Keeping Mobile Phone/Any Electronic Gadgets, Even in OFF position is treated as UFM
2. Don't Write Anything on Question Paper
3. Attempt All Questions
4. Assume Any Missing Data if required

SECTION-A (Maximum Knowledge Level: K2)

Q. No.	Question	CO	KL	PI
1. Attempt All Questions				(1 x 5)
(a)	Find the null space and range of linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(x_1, x_2) = (x_1 + x_2, x_1)$.	CO2	K2	2.4.1
(b)	Define Linear Transformation.	CO2	K1	2.4.1
(c)	Show that the linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by $T(x, y, z) = (x - y, y - z)$ is singular.	CO2	K2	2.4.2
(d)	For what value of k will the vector $(1, k, 5)$ in $V_3(\mathbb{R})$ be the linear combination of vectors $(1, 2, 0)$, $(0, -1, 1)$.	CO1	K2	2.1.3
(e)	Find the value of a , if the vectors $(1 + a, 1 - a)$ and $(1 - a, 1 + a)$ in $V_3(\mathbb{C})$ are linearly dependent.	CO1	K2	2.1.3

SECTION-B (Minimum Knowledge Level: K2)

2. Attempt Any two Question		(Same K Levels) (2 x 2)		
(a)	Show that the mapping $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T(x, y) = (2x - y, x - y, -2x)$ is linear transformation.	CO2	K2	2.4.1, 2.4.4
(b)	Show that the linear operator T on $\mathbb{R}^3$ is invertible and find T^{-1} , where $T(x, y, z) = (3x, x - y, 2x + y + z)$.	CO2	K3	2.4.2
(c)	$B = (1, 1 + x, 1 + x + x^2)$ and $B' = (2 + x + x^2, x + x^2, x)$ are the basis of vector space V , the set of all the polynomials of degree 2. Find the change of basis matrix from B to B' .	CO2	K2	2.1.3, 2.4.1

SECTION-C (Minimum Knowledge Level: K3)

3. Attempt Any one Question		(Same K Levels) (3 x 1)		
(a)	State Rank Nullity Theorem and verify it for a linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by, $T(x, y, z) = (x + 2y - z, y + z, x + y - 2z)$.	CO2	K3	2.4.1, 2.4.4

(b)	Find the matrix $[T]_{B,B'}$ of linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by $T(x, y, z) = (x + y, y + z)$ relative to the basis $B = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ and $B' = \{(1, 2), (2, 5)\}$. Also verify that $[T]_{B,B'} [v]_B = [T(v)]_{B'}$.	CO2	K3	2.1.3, 2.4.1
4. Attempt Any one Question (Same K Levels) (3 x 1)				
(a)	Show that the set $B = \{(2, 1, 4), (1, -1, 2), (3, 1, -2)\}$ form a basis of vector space $\mathbb{R}^3(\mathbb{R})$.	CO1		2.1.3
(b)	Find the basis and dimension of subspace W of vector space $\mathbb{R}^4(\mathbb{R})$ generated by the vectors $(1, 4, -1, 3), (2, 1, -3, -1), (0, 2, 1, 5)$ and extend the basis of W to find the basis of vector space $\mathbb{R}^4(\mathbb{R})$.	CO1		2.4.1, 2.4.2

CO Course Outcomes mapped with respective question
 KL Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)
 K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create



ABES ENGINEERING COLLEGE, GHAZIABAD (AUTONOMOUS)

Pre - End Term Exam (Session: 2025-26)

Admission No.:

Course Name: B.Tech

Subject Name: Fundamentals of Linear Algebra & Statistics

Subject Code: 25AS102

Max.Marks:20

Time: 90 Minutes

Semester: I

Note:

1. Keeping Mobile Phone/Any Electronic Gadgets, Even in OFF position is treated as UFM
2. Don't Write Anything on Question Paper
3. Attempt All Questions
4. Assume Any Missing Data if required

SECTION-A (Knowledge Level: K2 & K3)

SECTION-A(Knowledge Level: K2 &K3)																															
Q. No.	Question						CO	KL	PI																						
Attempt anyone part from each question						(3 Marks x 4 Questions = 12 Marks)																									
1.(a)	Find the arithmetic mean of the following data using assumed mean '25': <table><tr><td>Marks</td><td>0-10</td><td>10-20</td><td>20-30</td><td>30-40</td><td>40-50</td></tr><tr><td>No. of students</td><td>5</td><td>10</td><td>40</td><td>20</td><td>25</td></tr></table>						Marks	0-10	10-20	20-30	30-40	40-50	No. of students	5	10	40	20	25	CO4	K2	2.1.3										
Marks	0-10	10-20	20-30	30-40	40-50																										
No. of students	5	10	40	20	25																										
1.(b)	Find the median from the following table <table><tr><td>Wages in '000</td><td>0-10</td><td>10-20</td><td>20-30</td><td>30-40</td><td>40-50</td><td>50-60</td><td>60-70</td><td>70-80</td></tr><tr><td>No. of workers</td><td>2</td><td>18</td><td>30</td><td>45</td><td>35</td><td>20</td><td>6</td><td>3</td></tr></table>						Wages in '000	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	No. of workers	2	18	30	45	35	20	6	3	CO4	K2	2.1.3				
Wages in '000	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80																							
No. of workers	2	18	30	45	35	20	6	3																							
2.(a)	Find the coefficient of variation for the following data: <table><tr><td>Marks</td><td>1-3</td><td>3-5</td><td>5-7</td><td>7-9</td><td>9-11</td></tr><tr><td>No. of students</td><td>15</td><td>18</td><td>27</td><td>10</td><td>6</td></tr></table>						Marks	1-3	3-5	5-7	7-9	9-11	No. of students	15	18	27	10	6	CO4	K2	2.4.1										
Marks	1-3	3-5	5-7	7-9	9-11																										
No. of students	15	18	27	10	6																										
2.(b)	The number of patients seen in the emergency ward of a hospital for a sample of 5 days in the last month was 153, 147, 151, 156 and 153. Determine the coefficient of mean absolute deviation.						CO4	K3	2.4.1																						
3.(a)	Fit a second degree data for the following data: <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>y</td><td>6</td><td>11</td><td>18</td><td>27</td></tr></table>						x	1	2	3	4	y	6	11	18	27	CO5	K2	2.4.1												
x	1	2	3	4																											
y	6	11	18	27																											
3.(b)	Write the normal equations for the following curves: (i) $y = ax + \frac{b}{\sqrt{x}}$ (ii) $y = \frac{c_0}{x} + c_1x^2$ (iii) $y = a_1e^{-2t} + a_2e^{-5t}$						CO5	K2	2.4.2																						
4.(a)	Find the rank correlation coefficient from the following data: <table><tr><td>x</td><td>68</td><td>64</td><td>75</td><td>50</td><td>64</td><td>80</td><td>75</td><td>40</td><td>55</td><td>64</td></tr><tr><td>y</td><td>62</td><td>58</td><td>68</td><td>45</td><td>81</td><td>60</td><td>68</td><td>48</td><td>50</td><td>70</td></tr></table>						x	68	64	75	50	64	80	75	40	55	64	y	62	58	68	45	81	60	68	48	50	70	CO5	K2	2.4.1
x	68	64	75	50	64	80	75	40	55	64																					
y	62	58	68	45	81	60	68	48	50	70																					

4.(b)	In a partially destroyed laboratory record of an analysis of a correlation data, the following results only are legible: Variance of $x = 9$, Regression equations: $8x - 10y + 66 = 0$, $40x - 18y = 214$. What were (a) the mean values of x and y (b) the coefficient of correlation between x and y (c) the standard deviation of y	CO5	K3	2.4.1, 2.4.2																
SECTION-B (Knowledge Level: K3)																				
Attempt any one part from each question(4Marks x 2 Questions = 8 Marks)																				
5.(a)	Find the measure of kurtosis for the following distribution: <table><tr><td>Class</td><td>0-10</td><td>10-20</td><td>20-30</td><td>30-40</td><td>40-50</td><td>50-60</td><td>60-70</td></tr><tr><td>Frequency</td><td>8</td><td>12</td><td>20</td><td>30</td><td>15</td><td>10</td><td>5</td></tr></table>	Class	0-10	10-20	20-30	30-40	40-50	50-60	60-70	Frequency	8	12	20	30	15	10	5	CO4	K2	2.1.3
Class	0-10	10-20	20-30	30-40	40-50	50-60	60-70													
Frequency	8	12	20	30	15	10	5													
5.(b)	The first four moments of a distribution about $x = 2$ are 1, 16, -40 and 100 respectively. Calculate first four moments about mean and about origin.	CO4	K2	2.1.3																
6.(a)	Obtain a relation of the form $y = ae^{bx}$ for the following data by least square method: <table><tr><td>x</td><td>1</td><td>5</td><td>7</td><td>9</td><td>12</td></tr><tr><td>y</td><td>10</td><td>15</td><td>12</td><td>15</td><td>21</td></tr></table>	x	1	5	7	9	12	y	10	15	12	15	21	CO5	K2	2.4.1				
x	1	5	7	9	12															
y	10	15	12	15	21															
6.(b)	The following table gives the age (x) in years of cars and annual maintenance cost (y) in hundred rupees: <table><tr><td>x</td><td>1</td><td>3</td><td>5</td><td>7</td><td>19</td></tr><tr><td>y</td><td>15</td><td>18</td><td>21</td><td>23</td><td>22</td></tr></table> Estimate the maintenance cost for a 4 year old car after finding the regression equation.	x	1	3	5	7	19	y	15	18	21	23	22	CO5	K3	2.4.2				
x	1	3	5	7	19															
y	15	18	21	23	22															

CO Course Outcomes mapped with respective question
KL Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)
K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create



ABES ENGINEERING COLLEGE, GHAZIABAD (AUTONOMOUS)

Pre - End Term Exam (Session: 2025-26)

Admission No.:

Course Name: B.Tech

Subject Name: Fundamentals of Linear Algebra & Statistics

Subject Code: 25AS102

Max.Marks:20

Time: 90 Minutes

Semester: I

Note:

1. Keeping Mobile Phone/Any Electronic Gadgets, Even in OFF position is treated as UFM.
2. Don't Write Anything on Question Paper
3. Attempt All Questions
4. Assume Any Missing Data if required

SECTION-A (Knowledge Level: K2 & K3)

Q. No.		Question		CO	KL	PI																				
Attempt anyone part from each question		(3 Marks x4 Questions = 12 Marks)																								
1.(a)	Find the arithmetic mean of the following data using assumed mean '20':	CO4	K2	2.1.3																						
	<table><tr><td>x</td><td>5</td><td>10</td><td>15</td><td>20</td><td>25</td><td>30</td><td>35</td></tr><tr><td>f</td><td>8</td><td>15</td><td>20</td><td>32</td><td>23</td><td>17</td><td>5</td></tr></table>	x	5	10	15	20	25	30	35	f	8	15	20	32	23	17	5									
x	5	10	15	20	25	30	35																			
f	8	15	20	32	23	17	5																			
1.(b)	Calculate the mode from the following data:	CO4	K2	2.1.3																						
	<table><tr><td>Wages in '000</td><td>Below 100</td><td>100-200</td><td>200-300</td><td>300-400</td><td>400-500</td><td>Above 500</td></tr><tr><td>No. of workers</td><td>8</td><td>12</td><td>25</td><td>15</td><td>10</td><td>6</td></tr></table>	Wages in '000	Below 100	100-200	200-300	300-400	400-500	Above 500	No. of workers	8	12	25	15	10	6											
Wages in '000	Below 100	100-200	200-300	300-400	400-500	Above 500																				
No. of workers	8	12	25	15	10	6																				
2.(a)	Find the coefficient of variation for the following data:	CO4	K2	2.4.1																						
	<table><tr><td>Value</td><td>10-20</td><td>20-30</td><td>30-40</td><td>40-50</td><td>50-60</td><td>60-70</td><td>70-80</td></tr><tr><td>Frequency</td><td>1</td><td>20</td><td>69</td><td>108</td><td>78</td><td>22</td><td>2</td></tr></table>	Value	10-20	20-30	30-40	40-50	50-60	60-70	70-80	Frequency	1	20	69	108	78	22	2									
Value	10-20	20-30	30-40	40-50	50-60	60-70	70-80																			
Frequency	1	20	69	108	78	22	2																			
2.(b)	A software engineer recorded the execution time (in millisecond) of a program across 6 test runs: 120, 125, 118, 130, 122, 127. Determine the coefficient of mean absolute deviation to check the consistency of program execution.	CO4	K3	2.4.1																						
3.(a)	Fit a second degree data for the following data:	CO5	K2	2.4.1																						
	<table><tr><td>x</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>y</td><td>14</td><td>17</td><td>20</td><td>24</td><td>29</td></tr></table>	x	2	3	4	5	6	y	14	17	20	24	29													
x	2	3	4	5	6																					
y	14	17	20	24	29																					
3.(b)	Write the normal equations for the following curves:	CO5	K2	2.4.2																						
	$y = ax + \frac{b}{x}$ (ii) $y = \frac{a}{\sqrt{x}} + bx^2$ (iii) $y = a_1e^{3t} + a_2e^{-4t}$																									
4.(a)	Find the rank correlation coefficient from the following data:	CO5	K2	2.4.1																						
	<table><tr><td>x</td><td>23</td><td>27</td><td>28</td><td>28</td><td>29</td><td>30</td><td>31</td><td>33</td><td>35</td><td>36</td></tr><tr><td>y</td><td>18</td><td>20</td><td>22</td><td>27</td><td>21</td><td>29</td><td>27</td><td>29</td><td>28</td><td>29</td></tr></table>	x	23	27	28	28	29	30	31	33	35	36	y	18	20	22	27	21	29	27	29	28	29			
x	23	27	28	28	29	30	31	33	35	36																
y	18	20	22	27	21	29	27	29	28	29																

$$③ y = a_1 e^{(3t+4t)} + a_2$$

$$① y = a_1 e^{3t} + a_2 e^{-4t}$$

$$② y = a_1 e^{3t} + a_2 e^{-4t}$$

4.(b)	The two lines of regression for are given by: $x + 2y - 5 = 0$ and $2x + 3y - 8 = 0$ and variance of $x = 12$, What were (a) the mean values of x and y (b) the coefficient of correlation between x and y (c) the standard deviation of y	CO5	K3	2.4.1, 2.4.2
-------	---	-----	----	-----------------

SECTION-B (Knowledge Level: K3)

Attempt any one part from each question (4 Marks x 2 Questions = 8 Marks)

Find the measure of skewness for the following distribution:										CO4	K3	2.1.3
5.(a)	Value	6	12	18	24	30	36	42				
	Frequency	4	7	9	18	15	10	3				
The first four moments of a distribution about $x = 2$ are 1, 2.5, 5.5 and 16 respectively. Find first four moments about mean and about origin.										CO4	K2	2.1.3
6.(a)	Obtain a relation of the form $y = ab^x$ for the following data by least square method:									CO5	K3	2.4.1
	x	2	3	4	5	6						
	y	8.3	15.4	33.1	65.2	127.4						
6.(b)	Obtain the regression line of x on y for the data:									CO5	K3	2.4.2
	x	1	3	5	7	8	10					
	y	8	12	15	17	18	20					

CO Course Outcomes mapped with respective question

KL Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create